

Aegis Pulse: A Human-Centered AI Conceptual Framework for Exploring Bystander Trauma Care Design During the “Platinum Ten Minutes”

Mohamed Amdjed Dariadi (Team Lead — Project Strategy)

Sahil Indoria (Medical Researcher — Protocol Analysis)

Zeyad Bekhet (UI/UX Designer — Prototype Design)

Challenge Track: Human-Centered AI Challenge – NYAS Junior Academy

Team Number: #6678

Mentor: Dr. Daniel Lindegger

Spring 2026

Abstract

Every year, approximately 1.19 million people lose their lives due to road traffic crashes, with an estimated 92% of these fatalities occurring in low- and middle-income countries. Within pre-hospital care, the first ten minutes post-accident—often designated as the “Platinum Ten Minutes”—represent a vital window for victims experiencing urgent conditions, such as severe bleeding or temporary airway obstruction. Although bystanders are frequently present at accident scenes, psychological hesitation (the “freeze response”) and a lack of immediate direction often hinder early assistance.

This paper presents Aegis Pulse, an exploratory student-led conceptual framework designed for the New York Academy of Sciences Junior Academy Challenge. The project investigates a theoretical model combining conceptual device-native computer vision interfaces with simplified binary triage logic to illustrate how non-professional individuals might be guided through adaptive, audio-visual instruction concepts. Additionally, this study evaluates preliminary, self-reported survey feedback ($N = 50$) gathered across Algeria, Egypt, and the USA to explore public perception of pre-hospital challenges, concluding with a structural analysis of this academic design blueprint.

1 Introduction & Problem Definition

1.1 The Pre-Hospital Care Window

Data compiled by the World Health Organization (WHO) indicates that road traffic injuries remain a notable public health challenge, causing an estimated 1.19 million annual deaths and leaving numerous individuals with long-term disabilities. This ongoing global crisis creates significant socio-economic strains, affecting national resources and community well-being.

From a timeline perspective, trauma outcomes are often influenced by immediate actions. While critical injuries can cause immediate fatalities, secondary complications may lead to life-threatening conditions if untreated within the first hour, commonly referred to as the “Golden Hour.” Within this period, the initial ten minutes—the “Platinum Ten Minutes”—serve as an important phase where initial bystander stabilization concepts could offer localized support before professional Emergency Medical Services (EMS) reach the scene.

1.2 Contextual Factors in Early Trauma Situations

The motivation for exploring non-professional, interface-guided support concepts lies in the progressive nature of early trauma conditions:

- **Initial Hemorrhage Management:** Uncontrolled bleeding is widely cited as a major preventable factor in post-crash mortality. Public health campaigns like *Stop the Bleed* emphasize that a substantial portion of trauma victims require localized care before professional medical teams arrive, as severe blood loss may lead to critical complications if unaddressed within a brief timeframe.
- **Airway Position Management:** Positional complications or minor structural blockages can rapidly impact breathing in unconscious individuals. Since institutional EMS response times in many regions can average between 10 to 20 minutes, an open gap exists where conceptual bystander tool designs might offer immediate, basic positioning guidance.

1.3 The Psychological Dimension: The Bystander Freeze Response

A central theme of human-centered design exploration is recognizing that a bystander’s hesitation to assist at an emergency scene is frequently linked to psychological anxiety rather than a lack of empathy. In unexpected crisis environments, untrained individuals often experience a psychological “freeze response,” driven by fear of making mistakes or causing secondary harm. The Aegis Pulse framework is proposed as a theoretical design concept to explore how user interfaces can decouple a willingness to assist from specialized training, translating basic humanitarian steps into highly structured, accessible visual prompts.

1.4 Research Question

To guide this conceptual study and anchor its academic scope within the student innovation framework, we define the following core research question:

This study explores whether simplified, AI-guided conceptual decision flows and minimalist interface designs can reduce perceived hesitation and cognitive friction among untrained bystanders during early-stage emergency situations.

2 Proposed System Architecture & Conceptual Design

The theoretical architecture of the Aegis Pulse framework functions as a conceptual design model intended to minimize cognitive strain, mapping directly onto the user-interface interactive wireframes developed within Figma during this challenge track.

2.1 Component A: Conceptual Visual Assessment Blueprint

Standard software layouts often require complex text entry or search operations, which are structurally difficult to navigate under acute panic. The Aegis Pulse design model proposes an alternative visual approach. Upon launching the interface concept, the system is theoretically designed to utilize a smartphone’s camera feed. By exploring the potential future integration of edge-processed computer vision models, the framework illustrates how an interface could screen a scene to recognize basic visual markers—such as heavy blood pooling or unconscious postures—to automatically suggest a general emergency category without requiring manual text inputs from a user.

2.2 Component B: Binary Triage Logic Flow

Once a general context is isolated, the conceptual design channels user interaction through a simplified Binary Triage Logic model. Drawing inspiration from standard triaging workflows (such as the START protocol), this concept translates complex instructional paths into single, highly visible prompts within our Figma design that require only a binary response (**YES** or **NO**). For example, the prototype screen asks: *“Is the victim breathing normally?”* or *“Is there heavy pulsing blood?”* Limiting the design model to clear binary choices serves as a theoretical mechanism to bypass cognitive overload.

2.3 Component C: Localized Guidance Concepts

To account for low-connectivity environments, the framework suggests a conceptual model where core instructional assets run locally on the device’s hardware. Based on the binary user inputs, the user-centered interface concept is designed to render clear, localized audio-visual directions. These assets translate established humanitarian protocols into highly simplified tasks, such as: *“Apply continuous firm pressure to the wound using a clean cloth.”* Concurrently, the conceptual model proposes using device telemetry to automatically prepare localized coordinate data for dispatch integration to help shorten the professional response loop.

2.4 The Theoretical Logic Flowchart

Below is the conceptual logic flowchart of the proposed Aegis Pulse design, outlining how the theoretical framework would structure data processing and bystander interaction paths:

3 Data Collection, Survey Feedback, & Figma Design Lessons

3.1 Self-Reported Survey Explorations

To gather preliminary insights regarding public perception of the pre-hospital care gap and explore the relevance of our design choices, self-reported survey data ($N = 50$) were collected across Algeria, Egypt, and the United States using a 12-question Google Forms questionnaire. The received feedback indicates the following trends:

- **Perceived Knowledge Gap:** Approximately 82% of the surveyed respondents reported that they do not possess specific first-aid training or operational knowledge needed to manage severe bleeding scenarios at an accident scene. **Reported Care Hesitation:** A notable subset of the sample (approximately 42%) indicated that they had wanted to assist during a past real-world emergency but refrained from taking physical action because they lacked clear, confident direction on the correct steps.
- **Perceived Emergency Delays:** Approximately 93% of participants suggested that professional emergency arrivals in their local areas frequently exceed the 10-minute threshold. This localized feedback highlights the perceived value of developing bystander-focused assistance concepts.

3.2 Qualitative Design Feedback & Figma Prototype Development

During our design phase, we gathered qualitative feedback by sharing early wireframe concepts of our interface flows—specifically the interactive Scan and Triage frames designed in Figma—with peers and survey participants. This process highlighted two structural human-centered design priorities:

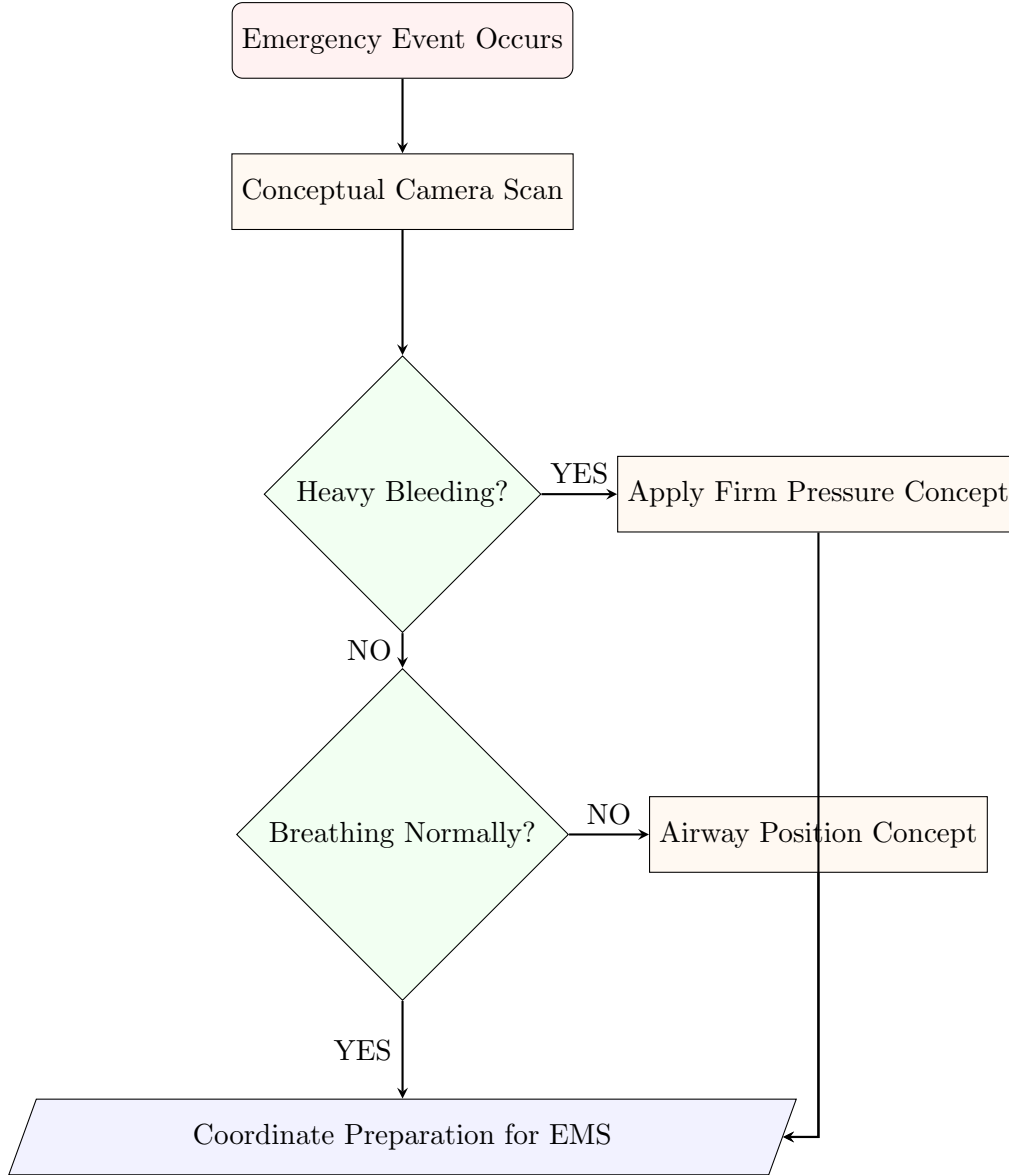


Figure 1: Proposed Aegis Pulse Conceptual Design Flowchart.

1. **Visual Optimization for Stress:** Feedback suggested that standard digital layouts and standard text blocks become difficult to process under emergency anxiety. In response, our Figma prototype layouts were altered to incorporate significantly enlarged fonts, high color contrasts, and minimal words per screen to preserve legibility.
2. **Background Action Integration:** Users noted that manually navigating away from instructions to call local emergency numbers creates an unnecessary operational burden. Consequently, the conceptual workflow was updated to theoretically initialize emergency services communication in the background the moment an active triage path is opened.

4 Research Scope & Limitations

To maintain proper academic accountability and scientific clarity, we define the boundaries and experimental limitations of this exploratory work:

- **Small Pilot Sample Size:** The analyzed dataset relies on a small-scale survey sample

($N = 50$). While useful for an initial exploratory study, these results represent subjective, self-reported opinions and cannot be generalized as a definitive global clinical or statistical representation.

- **absence of Functional Implementation:** This project is entirely a theoretical conceptual framework. No active software applications were built, no neural network models were coded or executed, and no live testing was performed on real-world accident scenes.
- **Design-Centric Focus:** The physical output of this study is strictly limited to static user-interface (UI/UX) mockups and flow designs constructed in Figma to serve as a visual conceptual exploration for a student innovation project.

5 Implementation Challenges & Strategic Model Pivot

5.1 Regulatory and Resource Context

Software systems that propose giving real-time, adaptive first-aid or diagnostic guidance are categorized as Software as a Medical Device (SaMD) by international health ministries like the FDA. Moving a conceptual framework into an actual public tool involves:

- **Multi-Year Clinical Evaluation:** The software logic and underlying computer vision concepts must go through extensive clinical testing to ensure that the provided guidance is entirely safe and never introduces secondary health risks.
- **Development and Capital Constraints:** Developing an official MedTech application involves navigating lengthy certification processes, complex data privacy laws (e.g., HIPAA), and substantial financial investments, which are beyond the capabilities of an independent student-led innovation group.

5.2 The Open-Access Academic Blueprint Direction

Recognizing these practical boundaries, our team has deliberately positioned Aegis Pulse as a **Conceptual Academic Research Blueprint** rather than a commercial deployment attempt. The value of this project is demonstrating how human-centered design patterns can be modeled to tackle public health literacy and bystander hesitation.

To expand this model in the future, we propose sharing this conceptual framework and our Figma design prototypes with larger institutional stakeholders, such as global humanitarian networks (e.g., Red Cross or Red Crescent societies), public health agencies, or academic research universities. These organizations have the necessary operational structures and clinical expertise to validate, govern, and distribute automated public safety tools safely.

6 Team Dynamics & Decentralized Collaboration

The development of Aegis Pulse represents an educational model for decentralized, multinational student collaboration across distinct geographic zones, connecting team members in Algeria (Medea), Egypt (Qena), and the United States (New Jersey).

6.1 The Module-Ownership Workflow

To balance our school commitments and work productively, we used a specialized “Module-Ownership” model. Each member held direct responsibility for an isolated area based on individual technical interests while maintaining an integrated collective workflow:

- **Mohamed Amdjed Dariadi** (Team Lead): Guided the project coordination, managed timeline milestones, and synchronized the research text and scientific synthesis.
- **Sahil Indoria** (Medical Researcher): Conducted the review of standard first-aid protocols, checked logic paths against international guidelines, and drafted the triage trees.
- **Zeyad Bekhet** (UI/UX Designer): Engineered the visual interface assets, design layouts, and interactive mobile wireframe prototypes inside Figma.

6.2 Asynchronous Synchronization

To navigate time-zone differences, our team scheduled bi-weekly video alignment sessions on weekends using scheduling tools to ensure absolute fairness. Daily communication was managed asynchronously via project messaging lines, following a shared 24-hour response policy that allowed members to participate around their local exam schedules without stalling team progress. Disagreements regarding design assets or research wording were resolved through discussion or consulting our challenge mentor, Dr. Daniel Lindegger, ensuring a supportive and collaborative environment throughout our participation in the Junior Academy.

7 Conclusion

The conceptual research compiled for Aegis Pulse demonstrates that an analytical, human-centered design approach can offer alternative strategies to study the care gap within the pre-hospital “Platinum Ten Minutes.” By combining smartphone camera concepts with basic binary choices, our conceptual layout illustrates a practical design path to help bystanders manage panic and assist effectively. While regulatory and financial factors keep a student project from current physical distribution, our project milestones successfully combine clinical guidelines, preliminary survey feedback, and user-centered design patterns into a cohesive, open-access blueprint for future public health exploration.

References

- American Heart Association. (2020). Part 8: First aid. *Circulation*, 142(suppl_2), S557–S579. <https://doi.org/10.1161/CIR.0000000000000897>
- American Red Cross. (2024). *First aid/CPR/AED guidelines and participant’s manual*. Red Cross Training Resources. <https://www.redcross.org>
- Department of Homeland Security. (2024). *Stop the Bleed Initiative*. <https://www.stopthebleed.org/>
- Association for the Advancement of Automotive Medicine. (2022). Traffic Injury Prevention, 23(4), 210–215. <https://doi.org/10.1080/15389588.2022.XXXXXXX>
- National Center for Biotechnology Information. (2023). *START triage system in mass casualty and trauma incidents*. National Library of Medicine. <https://www.ncbi.nlm.nih.gov/books/NBK395802/>
- The Lancet Global Health. (2022). Pre-hospital trauma care systems in low- and middle-income countries. *The Lancet Global Health*, 10(4), e568–e577. <https://www.thelancet.com/journals/langlo>
- World Health Organization. (2023). *Road traffic injuries fact sheet*. <https://www.who.int/news-room/fact-sheets/detail/road-traffic-injuries>

- World Health Organization. (2024). *Emergency medical team minimum data set (MDS) for trauma care*. <https://www.who.int/publications/i/item/9789240049123>
- Mayo Clinic. (2025). *First aid: Shock and hemorrhage management*. <https://www.mayoclinic.org/first-aid>